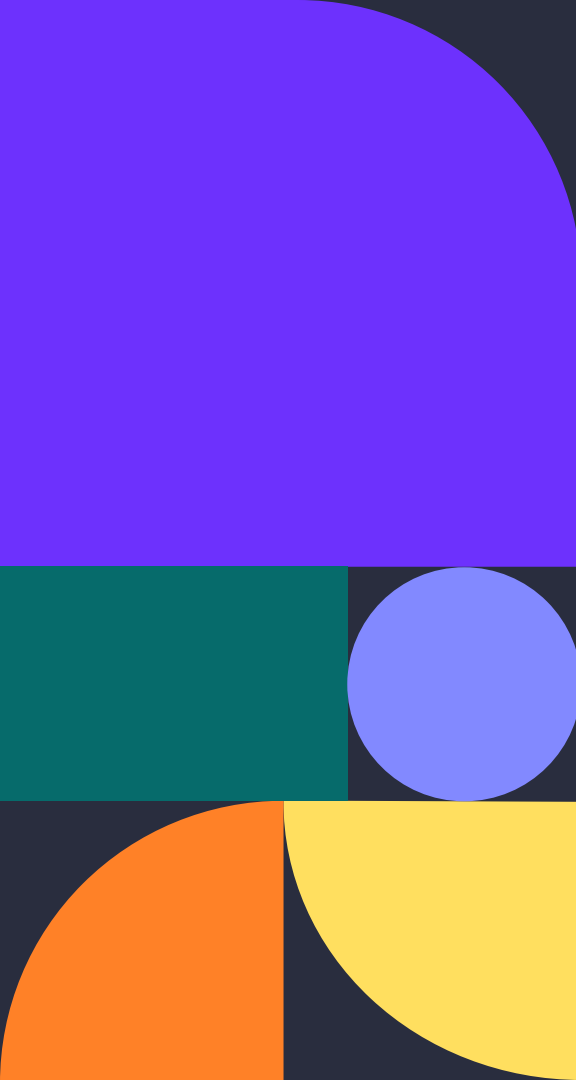




Noori (نوري):

Computational Modeling of Quranic Semantics Through Arabic NLP and Multi-Stream Embedding Systems

Ahmad Assaleh

05/20/2026

Computer Systems – Period 3 – Dr. Yilmaz

Problem

Our Core Challenge



Traditional Quranic study relies on human **interpretation, categorization** and **translation**

01. Interpretive Bias
among scholars

02. Dependency on
Translations

03. Hidden Semantic
Meanings

04. Limited Cross
Referencing

Existing Quran Search Systems use basic feature vectors and cannot discover the **multidimensional** nature of each verses



Section 1.

Arabic Background

Understanding the Arabic



Noun Form

"A Book"

کتاب

Simple

“He Wrote”

کتاب

The Quran uses complete diacritization (harakat) as well as the addition of specific letters to stem from a singular root word

Application In the Quran

One of the **BIGGEST** controversies in the Quran can be attributed to **Interpretive Bias**

Meaning?
Let's See...

ضَرَبَ

Modern Arabic:
-To Hit/Strike

Quranic Arabic:
-To Set Forth

Different FORMS in the Quran:

وَاضْرِبْ - And Present

فَاضْرِبْ - To Strike

ضَرْبًا - (to) Move About

وَضُرِبَتْ - And Struck



Section 2.

Motivation & Approach

Topic Based Categorization

Human imposed
categories

Very rigid
structure

Chat Bots (ChatGPT)

Can easily
hallucinate

Not trustworthy
in information
heavy texts

Search Engines (Google)

Only find exact
word matches

Translation
Dependant

Basic Semantic Search

English
dependant

Complete lack
morphological
awareness

Other solutions



The Future of Studying The Quran

The model has to look good (eventually):

- Real Time exploration of verse relationships
- Visual Proximity representing Semantic Similarity

Understanding the Quran:

- Have an initial base model with all verses represented
- Transform the base model based on user curiosity



Start Here



Section 3.

Understanding The Algorithms

Quran Corpus

The Raw Data

Instead of using only the Quran as my data,
I need a deeper representation of each word:

- Quran Corpus breaks down each word into its different grammatical patterns
- Gives me a lot more training data to work with

[Surah:Verse:Word:Section] (105:2:5:1)

taḍlīlin (taDoliylK)

Raw:

STEM|POS:N|VN|(II)|LEM:taDoliyl|ROOT:DII|M|INDEF|GEN

Breakdown

POS: N → Noun

ROOT: ḍ-l-l (ض ل ل) → to go astray

VN → Verbal noun (مصدر / masdar)

(II) → Form II verb origin

[Surah:Verse:Word:Section] (113:4:3:2)

n~affātī (n~af~avati)

Raw:

STEM|POS:N|LEM:n~af~avat|ROOT:nfv|FP|GEN

Breakdown

POS: N → Noun

ROOT: n-f-v (ن ف ث) → to blow/spit lightly

FP → Feminine plural

GEN → Genitive case

PPMI and SVD For Root Analysis

Update matrix when a
root appears within ± 15
words of another root

~2,000 Roots Extracted From the Quran

The Root
Matrix



+0.5	+0.0	...	...	+1.7
+0.0	...	:	:	:
:	:	...	:	:
:	:	:	...	:
+1.7	...	...	...	+0.0

~2,000
Roots
Extracted
From the
Quran

PPMI and SVD For Root Analysis

$$\begin{bmatrix} +0.5 & +0.0 & \dots & \dots & +1.7 \\ +0.0 & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ +1.7 & \dots & \dots & \dots & +0.0 \end{bmatrix}$$

**The Root
Matrix**

~2,000 x ~2,000

SVD looks for
important
features and
simplifies to
512 ROWS



$$\begin{bmatrix} +0.5 & +0.0 & \dots & \dots & +1.7 \\ +0.0 & \ddots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ +1.7 & \dots & \dots & \dots & +0.0 \end{bmatrix}$$

**SVD Root
Matrix**

512 x ~2,000

Take every root, from every word, from every verse...
And **combine** them to make **ROOT FEATURE MATRIX** per verse

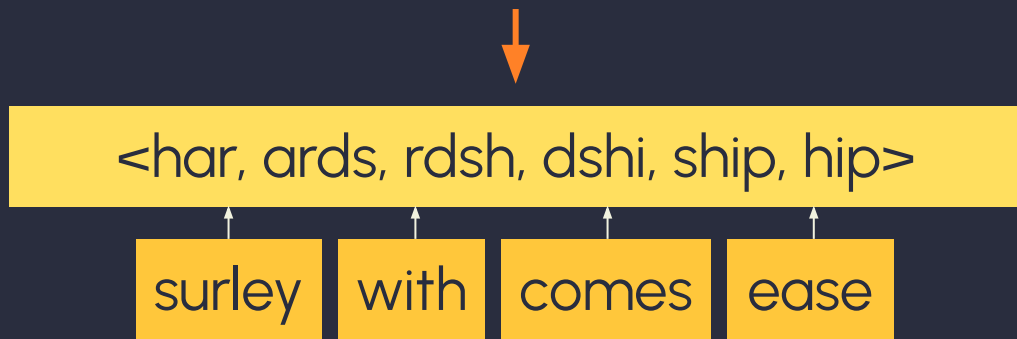
This new matrix will be 512 (Feature Dimensions) x 6,236 (Verses)



FastText Algorithms POV

Utilizing **surrounding** words to make sense of the “target word” and give it an **abstract** definition





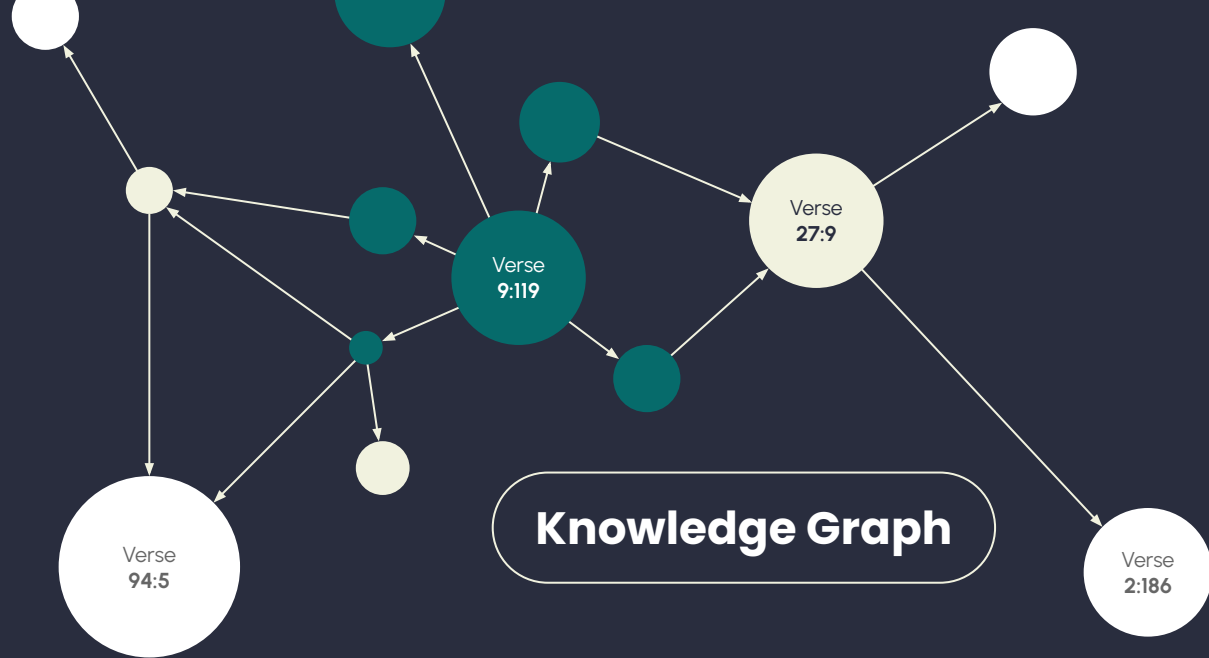
This is done over **millions** of rounds of training, over every single word, and every single word section

Target Word

Break it Down

Train each
section as a
part of
the whole

FastText Algorithm



Node2Vec Algorithm

Finding the relationships
between verses

Building the Knowledge Graph:

- Finding closely related verses based on proximity or word relations

Walking through the Verses:

- **Traversing the Graph:** Biased random walks based on the graph
- **Walk Paths:** Start with immediate neighbors, then slowly expand
- **Create Matrix:** Mapping commonly occurring verses together

512 Word
Dimensions

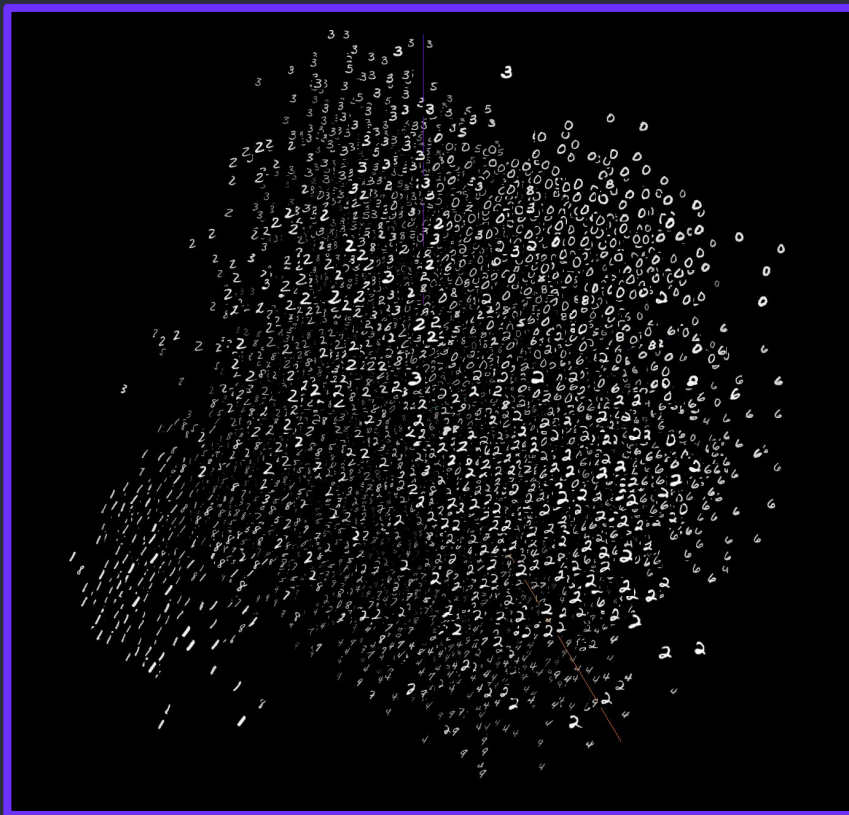
512 Verse
Dimensions

512 Root
Dimensions

Verse 2286	Verse 94.5	Verse 2255	Verse 3139
$\begin{bmatrix} -0.5 \\ -2.9 \\ -7.9 \\ -4.0 \\ 2.9 \\ 2.6 \\ -1.4 \\ 2.8 \\ \dots \\ 3.8 \end{bmatrix}$	$\begin{bmatrix} 2.1 \\ 9.0 \\ -1.5 \\ -4.0 \\ 2.9 \\ 2.5 \\ 7.0 \\ 3.6 \\ \dots \\ -2.8 \end{bmatrix}$	$\begin{bmatrix} -0.5 \\ -2.9 \\ -4.8 \\ -4.4 \\ -6.2 \\ 0.2 \\ 3.2 \\ 1.3 \\ \dots \\ 9.2 \end{bmatrix}$	$\begin{bmatrix} 1.5 \\ -1.8 \\ -6.2 \\ 2.3 \\ 9.0 \\ -1.2 \\ -1.4 \\ 2.8 \\ \dots \\ 3.8 \end{bmatrix}$

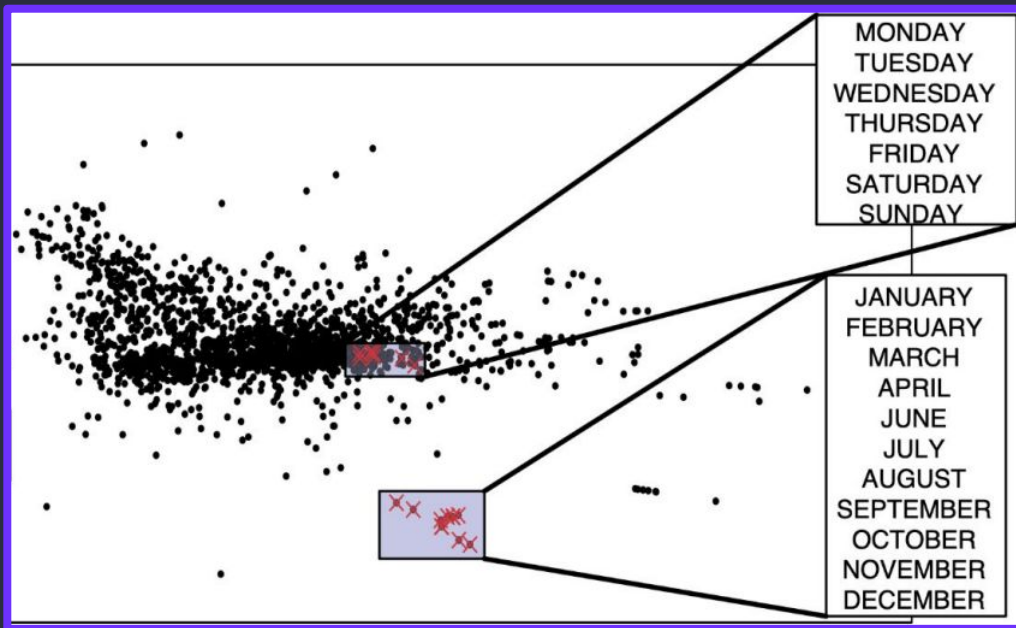
Compile all 3 into
embedding matrix

Spectral Graph Theory Visualization



What a
multi-dimensional
model COULD be
imagined to look

Spectral Graph Theory Visualization



What a
multi-dimensional
model COULD be
imagined to look



Section 4.

Algorithm Structure

Embedding

Quran

Verses &
Embeddings

NLP

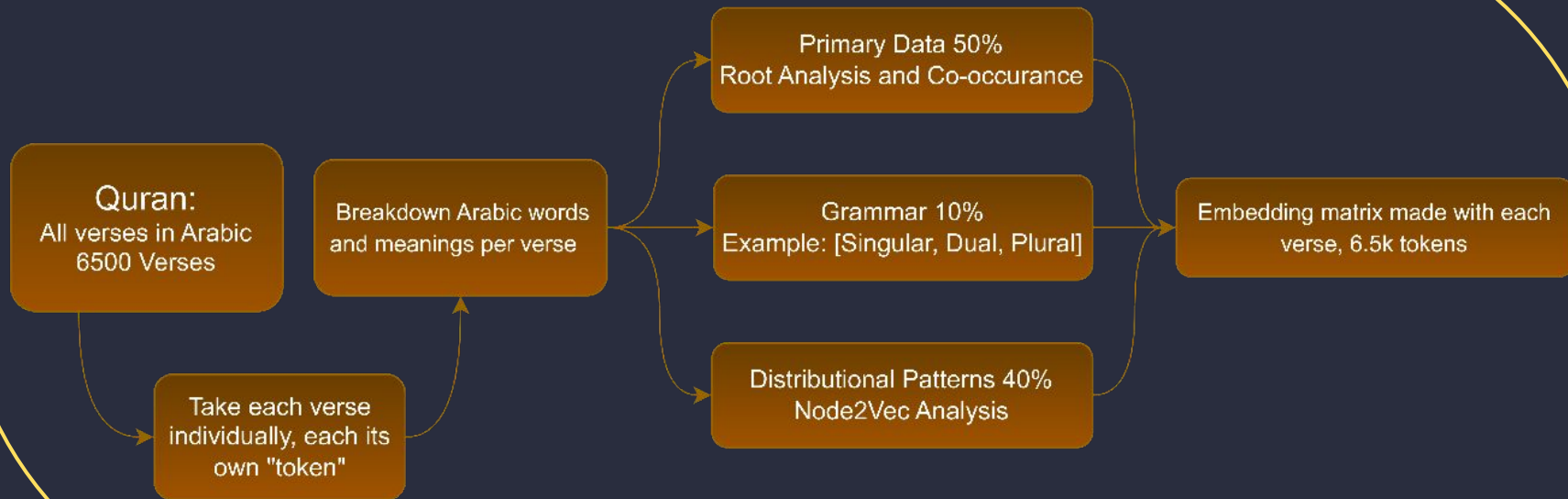
User
Prompt

Get contextual
understanding

Transformed
Verses

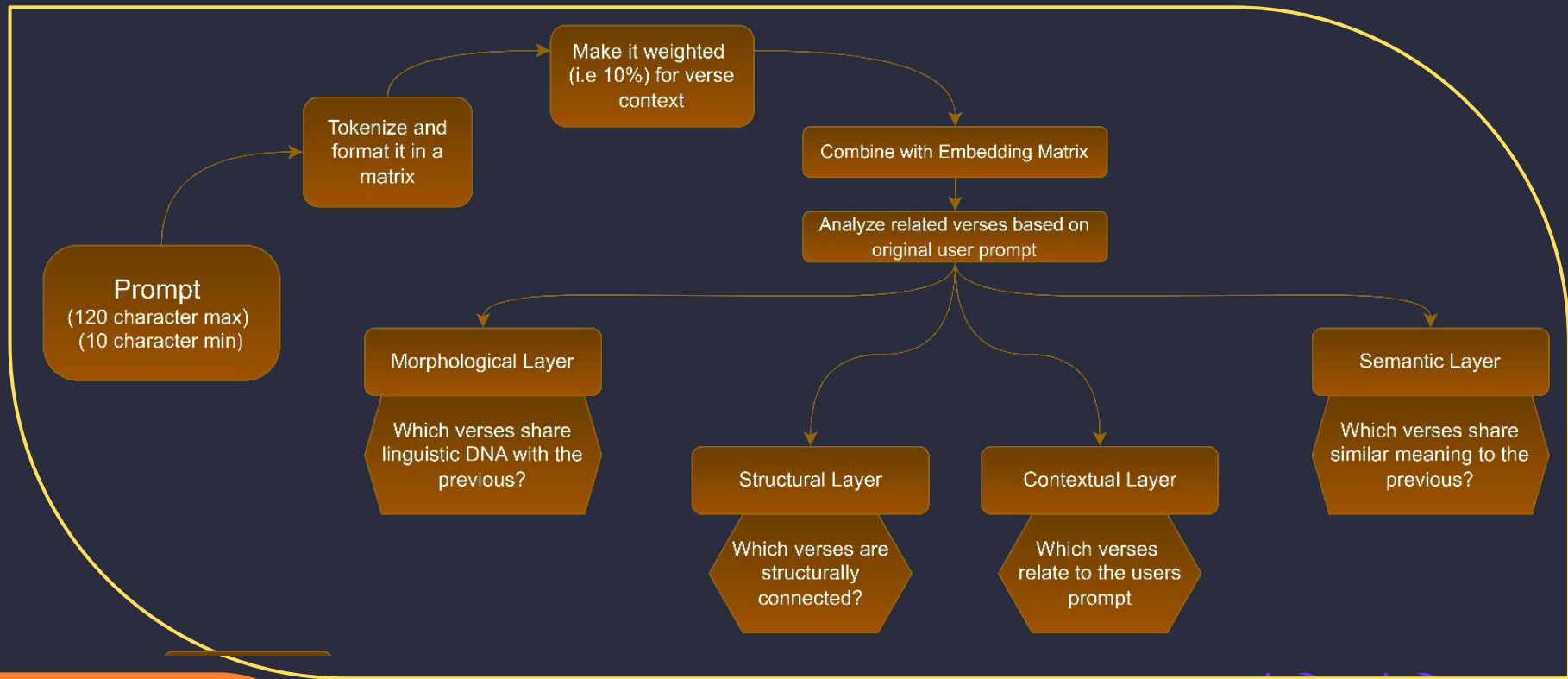
Start Building
Noori

Method Overview



Methods Section 1

Creating the
Embedding Matrix



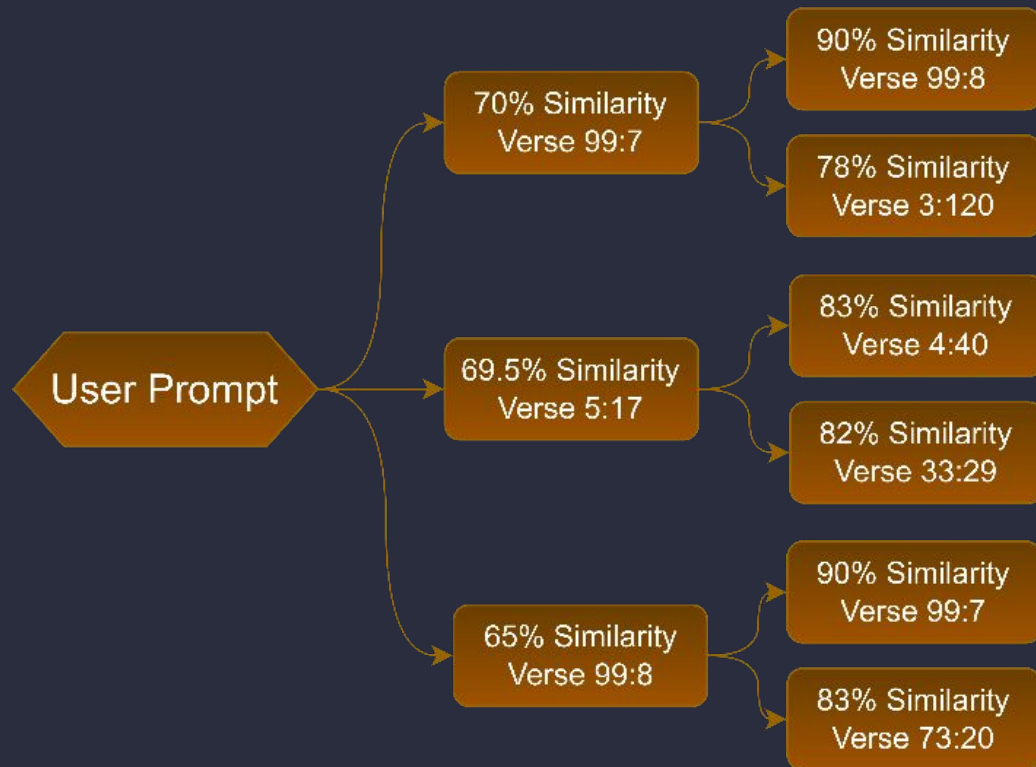
Methods

Section 2

Natural Language
Processing

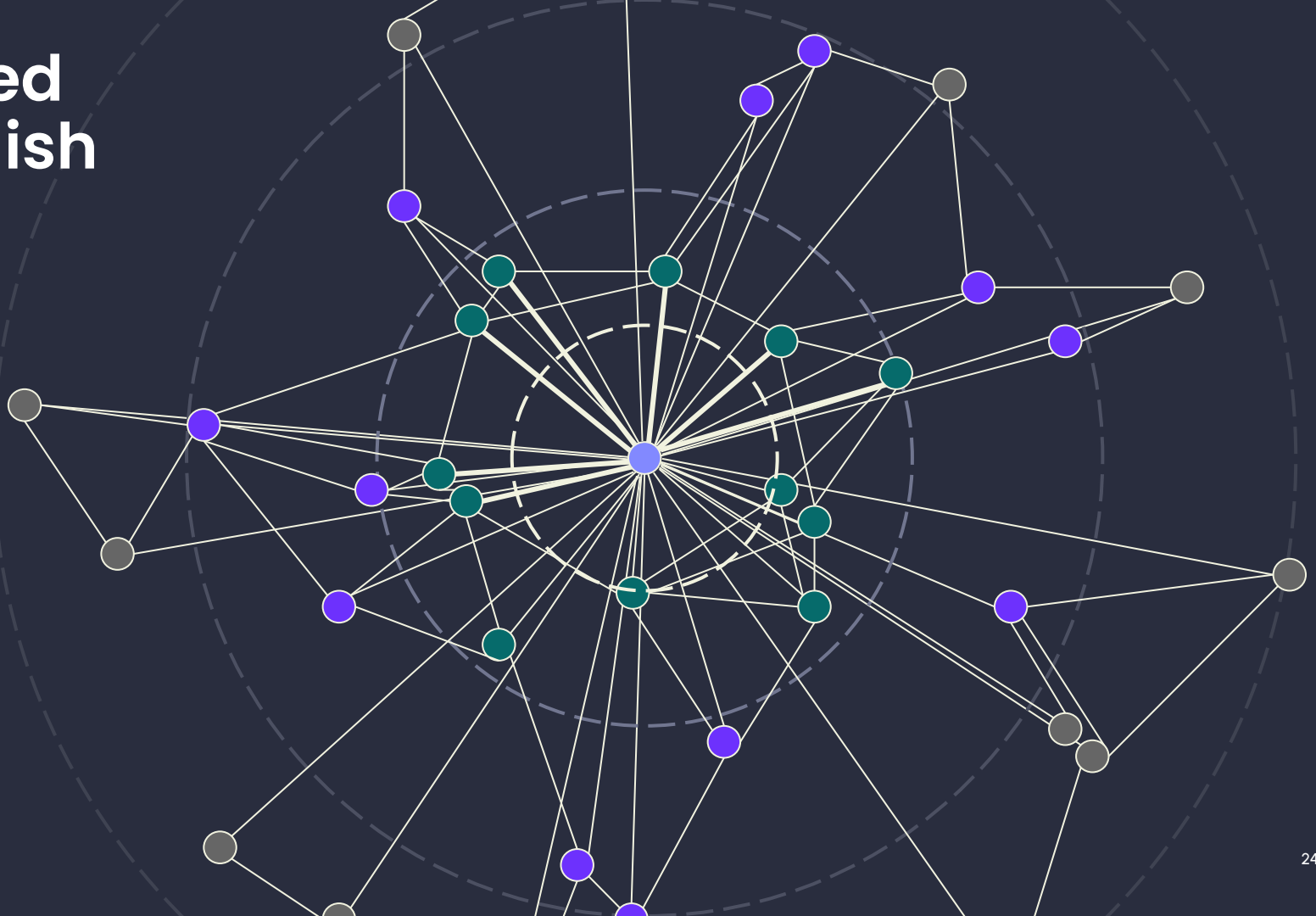
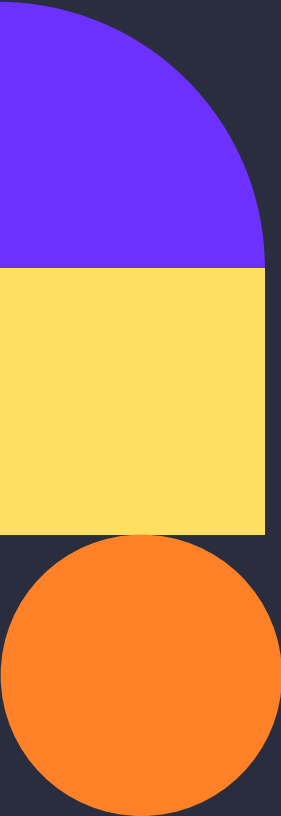
Final Step

Use context to predict the **"Next Verses"** based on a probability curve of possible verses



Methods Section 3

Expected Result-ish



Results Analysis

Verses related to
MERCY
and why the code
made the relation

Top 20 verses related to MERCY:

1. 73:20	score: 0.898	roots: rHm, gfr
2. 2:109	score: 0.898	roots: Efw
3. 9:99	score: 0.898	roots: rHm, gfr
4. 3:19	score: 0.890	roots: related (no direct root)
5. 4:94	score: 0.890	roots: related (no direct root)
6. 6:54	score: 0.889	roots: rHm, gfr
7. 40:7	score: 0.887	roots: rHm, gfr
8. 2:143	score: 0.879	roots: rHm
9. 57:21	score: 0.879	roots: gfr
10. 2:105	score: 0.878	roots: rHm
11. 74:31	score: 0.877	roots: related (no direct root)
12. 4:152	score: 0.877	roots: rHm, gfr
13. 60:4	score: 0.876	roots: gfr
14. 4:64	score: 0.876	roots: rHm, gfr
15. 7:156	score: 0.874	roots: rHm

Results Analysis

9:99 ▶ 📖



وَمِنَ الْأَعْرَابِ مَنْ يُؤْمِنُ بِاللَّهِ وَالْيَوْمِ الْآخِرِ وَيَتَّخِذُ مَا يُنْفِقُ قُرْبَتٍ عِنْدَ اللَّهِ
وَصَلَوَاتِ الرَّسُولِ أَلَا إِنَّهَا قُرْبَةٌ لَهُمْ سَيُدْخِلُهُمُ اللَّهُ فِي رَحْمَتِهِ إِنَّ اللَّهَ غَفُورٌ رَحِيمٌ ﴿٩٩﴾

However, among the nomadic Arabs are those who believe in Allah and the Last Day, and consider what they donate as a means of coming closer to Allah and 'receiving' the prayers of the Messenger. It will certainly bring them closer. Allah will admit them into His mercy. Surely Allah is All-Forgiving, Most Merciful.

89.8% Relation

Roots: rHm, gfr

Results Analysis

4:94 ▶ 📖



يَا أَيُّهَا الَّذِينَ ءَامَنُوا إِذَا ضَرَبْتُمْ فِي سَبِيلِ اللَّهِ فَتَبَيَّنُوا وَلَا تَقُولُوا لِمَنْ أَلْقَىٰ إِلَيْكُمُ السَّلَامَ
لَسْتَ مُؤْمِنًا تَبْتَغُونَ عَرَضَ الْحَيَاةِ الدُّنْيَا فَعِنْدَ اللَّهِ مَغَانِمُ كَثِيرَةٌ كَذَٰلِكَ
كُنْتُمْ مِن قَبْلُ فَمَنْ بَرَّ اللَّهُ عَلَيْكُمْ فَتَبَيَّنُوا إِنَّ اللَّهَ كَانَ بِمَا تَعْمَلُونَ خَبِيرًا



O believers! When you struggle in the cause of Allah, be sure of who you fight. And do not say to those who offer you 'greetings of' peace, "You are no believer!"—seeking a fleeting worldly gain.¹ Instead, Allah has infinite bounties 'in store'. You were initially like them then Allah blessed you 'with Islam'. So be sure! Indeed, Allah is All-Aware of what you do.

89.0% Relation

No Direct Roots

Results Analysis

Top 10 verses that are related to verse 4:40

Top 10 most similar verses:

1.	4:67		similarity: 0.805
2.	29:27		similarity: 0.782
3.	99:7		similarity: 0.773
4.	18:2		similarity: 0.766
5.	16:41		similarity: 0.760
6.	33:29		similarity: 0.752
7.	4:9		similarity: 0.749
8.	7:156		similarity: 0.748
9.	42:23		similarity: 0.744
10.	39:10		similarity: 0.744

Results Analysis



إِنَّ اللَّهَ لَا يَظْلِمُ مِثْقَالَ ذَرَّةٍ وَإِنْ تَكُ حَسَنَةً يُضَاعِفْهَا وَيُؤْتِ مِنْ لَدُنْهُ أَجْرًا عَظِيمًا

Indeed, Allah never wrongs ‘anyone’—even by an atom’s weight.¹ And if it is a good deed, He will multiply it many times over and will give a great reward out of His grace.

100% Relation with Verse 4:40

Results Analysis



إِنَّ اللَّهَ لَا يَظْلِمُ مِثْقَالَ ذَرَّةٍ وَإِنْ تَكُ حَسَنَةً يُضَاعِفْهَا وَيُؤْتِ مِنْ لَدُنْهُ أَجْرًا عَظِيمًا

Indeed, Allah never wrongs "anyone"—even by an atom's weight.¹ And if it is a good deed, He will multiply it many times over and will give a great reward out of His grace.



وَإِذَا لَآتَيْنَهُمْ مِنْ لَدُنَّا أَجْرًا عَظِيمًا

and We would have granted them a great reward by Our grace

80.5% Relation with Verse 4:40

Results Analysis

﴿٤٠﴾ إِنَّ اللَّهَ لَا يَظْلِمُ مِثْقَالَ ذَرَّةٍ وَإِنْ تَكُ حَسَنَةً يُضْعِفْهَا وَيُؤْتِ مِنْ لَدُنْهُ أَجْرًا عَظِيمًا

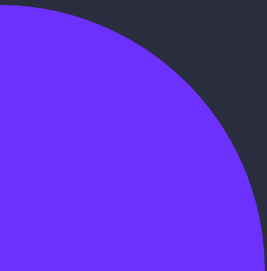
Indeed, Allah never wrongs "anyone"—even by an atom's weight.¹ And if it is a good deed, He will multiply it many times over and will give a great reward out of His grace.

﴿٧﴾ فَمَنْ يَعْمَلْ مِثْقَالَ ذَرَّةٍ خَيْرًا يَرَهُ

So whoever does an atom's weight of good will see it.

78.2% Relation with Verse 4:40

Current Demo



Browser tabs: New Tab, noori, screen recording windows - Go

Address bar: localhost:5173

Navigation bar: EDIT AI, College, MSA, Gmail, Scholarships, Pre and Planning, NLP and LLM's, Red Day HW, Blue Day HW, Jobs

**NOORI**
QURANIC ENLIGHTENMENT

Node size
reference-to-prompt
Edge weight
verse-to-verse similarity

Ask a question about the Quran, e.g. "What does Islam say about charity?" (press Enter)

EXPLORE

نور

ENTER A QUESTION TO ILLUMINATE THE QURAN

Ex: "How should we treat our guests?" "What is the purpose of prayer?"



Section 5.

Final Thoughts & Conclusion

Limitations

It is what it is...



The **biggest limitation** to my project currently is the underdeveloped **Arabic Sector** of AI

Limited **Mathematical** and **Quranic Knowledge** lead to →
Simpler models than expected

Other **BIG** limitations for this type of project

Time and Computation!!
Resorted to TJ servers in order to train my model

What does this mean for my project?

Not as much **functionality** on interactive website and **more training always better**

Impact

For Islamic Scholarship

Discover Hidden Connections

Create parallels with structures and morphological meanings

Cross-Reference Enhancement

Can map and elaborate on the chronological assortment of verses

For Students and Educators

Personalized Learning Paths

Students can explore the Quran without predefined curricula

Visual Understanding

Network visual will make abstract semantic relationships explorable

For Computational World

New Learning Potential

Apply to Classical Arabic Poetry and historic Islamic manuscripts

Benchmark for Arabic NLP

Now have a multi-modal attention architecture for Classic Arabic

Most importantly, this could serve as a **baseline** in which other researchers could use to **explore** any type of **text** with **minimal data**

Conclusion And Future Work

*The future belongs to those who **believe in the beauty of their dreams** – Eleanor Roosevelt*

01. Get in contact with experts in the field

02. Get an abstract and interactive site

03. Constantly update the algorithm

04. Expand to Quran school everywhere

Existing Quran Search Systems use basic feature searches and cannot discover the **multidimensional** nature of each verses, and **Noori** will not only expand on the **infinite wisdom** of the Quran, but will give us a means of capturing a fraction of it,
Telling a more complete story.

**Thank you!!
Any Questions?**